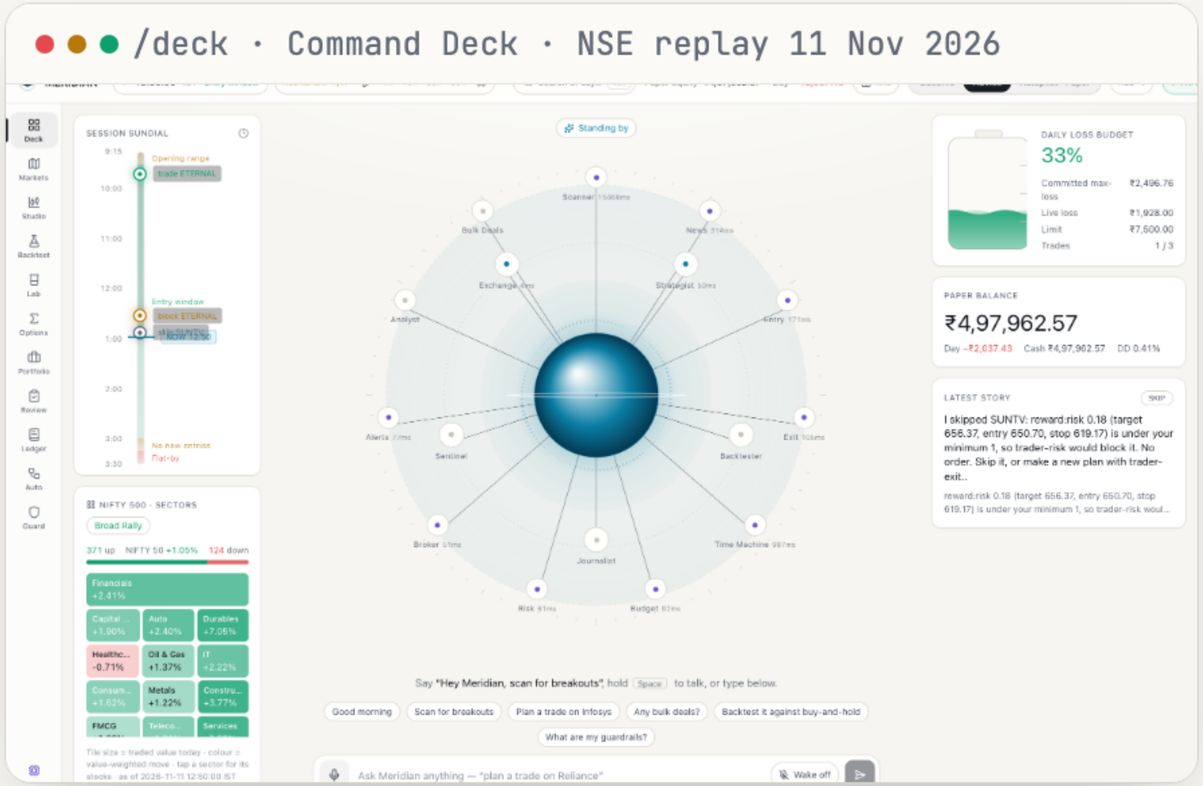
PS-02 · AGENTIC TRADING BOT · PAPER ONLY

MERIDIAN

Speak the market.
Trust every trade.

A voice-first paper-trading desk for India's NSE. One Conductor listens and drives sixteen agents; five defence layers stand between every idea and every order, and every decision gets a plain-English story.

16	5	497	₹0
agents	defence layers	NSE stocks	real money



"Hey Meridian, scan for breakouts"

"Zomato mein trade plan karo"

"Stop everything"

EVERY PS-02 REQUIREMENT, MET AND DEMONSTRABLE

- ✓ Reads data and news
- ✓ Picks a strategy
- ✓ Backtests it
- ✓ Paper account
- ✓ Sizes and stops
- ✓ Daily loss limit
- ✓ Plain-language journal
- ✓ Paper only
- ✓ Kill switch
- ✓ vs buy-and-hold

Traders know the rules. Nothing makes them follow them.

WHAT SEBI FOUND

7 in 10

individual intraday equity traders lost money in FY23 (SEBI study, July 2024)

93%

of individual F&O traders made losses over FY22–FY24 (SEBI study, Sept 2024)

Root causes: entries on noise, rules kept in the head, size set by emotion, revenge trades after a loss, and no record of *why*.

Moment	Before: a typical retail day	After: with Meridian
Finding a trade	Tips, charts, FOMO	Scanner ranks 497 NSE stocks on one rule
Sizing	"Feels right"	Budget sizes to a ₹2,500 max loss
Stop-loss	Often skipped or moved	Required before entry, OCO bracket
After a loss	Revenge re-entry	Tilt guard: 15-minute cool-down
Limits	Raised mid-day	Voice can only tighten them
Afterwards	No idea why	A Story and a post-mortem per trade

You talk
English or Hinglish, voice or ⌘K

Agents work
Scan, plan, test, size, gate

Rules hold
Code, not willpower

Sources: SEBI studies on intraday (Jul 2024) and F&O (Sep 2024) trader outcomes

One intelligence, sixteen tools, policy outside the model

Experience — Landing · Deck · Markets · Studio · Lab · Options · Review · Guardrails

Voice gateway — Wake word · VAD · Whisper STT · Hinglish · TTS

Conductor — intent grammar → planner → narrator

- Policy engine
- Tilt guard
- Automations
- Compliance

11 skill agents

Scanner · News · Entry · Exit · Time Machine · Budget · Risk · Broker · Alerts · Analyst · Bulk Deals

5 native services

Strategist · Backtester · Journalist · Sentinel · Paper Exchange

State and data — SQLite events · hash-chained audit · real NSE replay · Yahoo .NS

TECH STACK

Layer	Built with
Frontend	Next.js 16, React 19, Tailwind 4
Motion, 3D	Motion 12, three.js
Charts	lightweight-charts, SVG
Backend	Python 3.12, FastAPI, WebSocket
Agents	day-trader-skill, India edition
Voice	faster-whisper, OmniVoice
LLM	Local Ollama (qwen2.5), optional
Storage	SQLite WAL, SHA-256 chain
Reports	ReportLab PDFs
Testing	pytest, Playwright

The LLM has no order-placing tool. Every order passes the Policy engine and a fingerprinted Risk PASS.

The guardrails held on a real NSE session

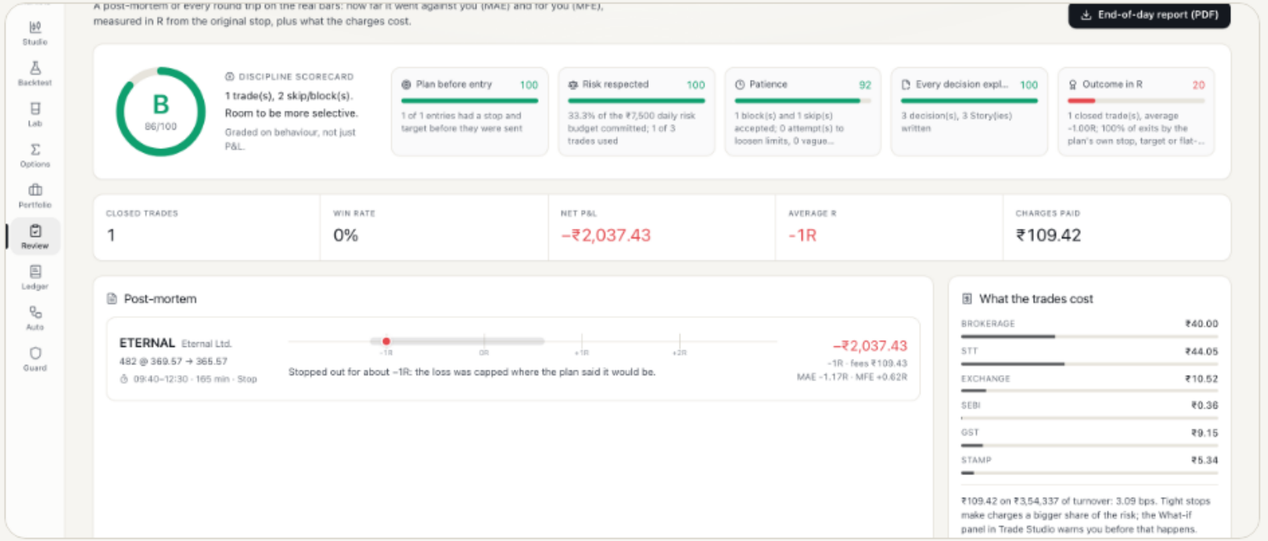
Risk on one ETERNAL trade, before and after

"Buy 600 Zomato", as asked — ₹3,108 · BLOCKED

482 shares, sized by Budget — ₹2,497 · PASS

Revenge re-entry after the stop — ₹0 · cool-down

Bar = max loss at the stop. Cap ₹2,500 a trade, ₹7,500 a day.



117 unit tests pass

25/25 end-to-end checks

100% decisions explained

Learners
Practise with real data and no money at risk

Busy traders
Hands-free scans, automations, alerts

Brokers, colleges
A safe sandbox with a SEBI-style audit trail

-1.17R worst point (MAE)

+0.62R best point (MFE)

₹109 charges, 3.1 bps

B · 86 discipline grade

Prototype figures, recorded NSE session; ₹ includes brokerage, STT, exchange, SEBI, GST, stamp duty.

Runs on a laptop today. Ready for a broker tomorrow.

TECHNICAL

Built and tested
13 screens, 50+ API routes, 117 tests, 25-check replay

Laptop CPU
Local STT, optional local LLM, no GPU needed

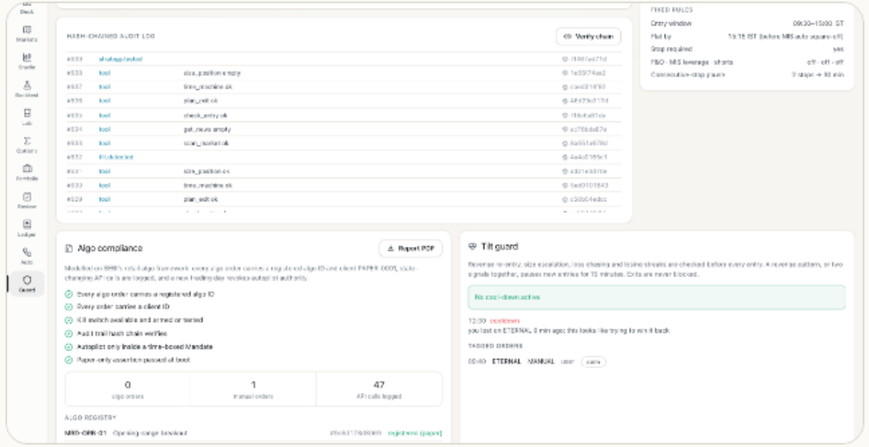
Real data
Recorded NSE session; live Yahoo .NS mode

ECONOMIC AND REGULATORY

Near-zero run cost
MIT and Apache-2.0 parts, free data, free local LLM

SEBI-aligned
Algo IDs, kill switch, API audit log, daily logout

Revenue paths
Free for learners; licensed to brokers and colleges



Risk	Mitigation
Unofficial data	Replay of a recorded day
LLM hallucination	Numbers validated vs tools
Real-money misuse	No live-broker code path

DEPLOY AND RUN COST
Web on **Vercel** · core in one **Docker** container

≈ ₹0 on free tiers

NOW	NEXT · 1 MONTH	3 MONTHS	6 MONTHS
Replay, voice, 16 agents	Live data, Hindi STT, PWA	Options chain, broker sync	Broker or college pilot

Paper only by design: no broker keys accepted anywhere.

Where Meridian sits, and what it is built on

Capability	Meridian	No-code algo	Options tools	Journals
Voice, English + Hinglish	Yes	No	No	No
Rules enforced before entry	Yes	Partly	No	No
Tilt guard with cool-down	Yes	No	No	Flags only
Story for every decision	Yes	No	No	Manual
Honest backtest vs hold	Yes	Yes	Partly	No
Algo IDs + audit chain	Yes	Broker-side	No	No

Category-level, from public descriptions (Streak, Sensibull, Edgewonk).

REFERENCES

SEBI · intraday trader study (Jul 2024); F&O study (Sep 2024)

SEBI · retail algo trading circular (4 Feb 2025)

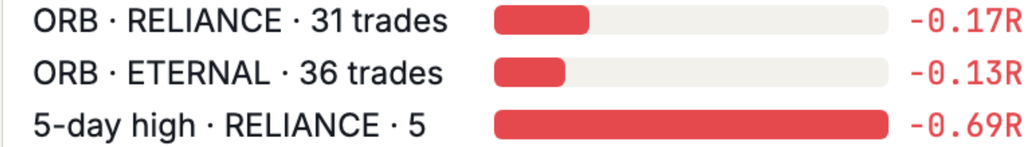
day-trader-skill · 11 trading skills (MIT)

k2-fsa OmniVoice · multilingual TTS (Apache-2.0)

faster-whisper · local speech recognition

Data · NSE India archives, Yahoo Finance, Google News

STRATEGY LAB · EXPECTANCY AFTER COSTS (R)



All three rejected by the edge gate: honest verdicts, not hype.

FROM RESEARCH TO DESIGN DECISIONS

Most retail traders lose money
→ **risk gate on every order**

Journals flag tilt after the fact
→ **tilt guard enforces it before**

SEBI algo rules: IDs, kill switch
→ **algo IDs, audit log, report**

India trades in many languages
→ **Hinglish voice and replies**

Educational prototype. Paper trading only. Not investment advice.